

1) Sea $T: M_{2 \times 2}$ la T.L. definida por:

$$T\left(\begin{bmatrix} x & y \\ z & w \end{bmatrix}\right) = \begin{bmatrix} y & w-x \\ 0 & z \end{bmatrix}$$

a) Calcule $T\left(\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\right)$ y $T\left(\begin{bmatrix} 3 & -4 \\ 4 & 5 \end{bmatrix}\right)$

b) Calcule $N(t)$; $V(t)$; $\rho(t)$

a) $x=1$, $y=1$, $z=0$, $w=1$

$$T\left(\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\right) = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$x=3$, $y=-4$, $z=4$, $w=5$

$$T\left(\begin{bmatrix} 3 & -4 \\ 4 & 5 \end{bmatrix}\right) = \begin{bmatrix} -4 & 2 \\ 0 & 4 \end{bmatrix}$$

b) $\begin{bmatrix} y & w-x \\ 0 & z \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

$$y=0 \quad w=x \quad 0=0 \quad z=0$$

$$N(t) = \begin{bmatrix} 0 & x \\ x & 0 \end{bmatrix} = x \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \Rightarrow \text{Gen} \left\{ \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right\}$$

$$V(t) = 1$$

$$\text{rango} = 3$$

2) Sean $B_1 = \{u_1 = (1, 1); u_2 = (0, 1)\}$ y $B_2 = \{v_1 = (1, 0, 0); v_2 = (1, 1, 0); v_3 = (1, 1, 1)\}$ son las bases ordenadas de $\mathbb{R}^2$ y $\mathbb{R}^3$. Si $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ es la T.L. tal que $T(1, 1) = (1, -1, 2)$ y $T(0, 1) = (3, 0, -1)$

Si sabemos que:

$$\begin{cases} T\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = (1, -1, 2) \\ T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = (3, 0, -1) \end{cases}$$

$$(x, y) = a(1, 1) + b(0, 1)$$

$$\begin{cases} a = x \\ a + b = y \end{cases} \quad b = y - x$$

$$([x, y]) = T(x(1, 1) + (y-x)(0, 1))$$

$$([x, y]) = xT(1, 1) + (y-x)T(0, 1)$$

$$([x, y]) = x(1, -1, 2) + (y-x)(3, 0, -1)$$

$$([x, y]) = \begin{bmatrix} x + 3y - 3x \\ -x \\ 2x - y + x \end{bmatrix} = \begin{bmatrix} -2x + 3y \\ -x \\ 3x - y \end{bmatrix} = \begin{bmatrix} -2 & 3 \\ -1 & 0 \\ 3 & -1 \end{bmatrix}$$

b) Calcule $[T]_{B_2 B_1}$ y $[T(u)]_{B_2}$ donde $u = 3u_1 + 4u_2$

$$\left[\begin{array}{cc|cc} 1 & 1 & 1 & 3 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 2 & -1 \end{array} \right] \xrightarrow{-R_2 + R_1} \left[\begin{array}{cc|cc} 1 & 0 & 0 & 2 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 2 & -1 \end{array} \right] \xrightarrow{-\frac{1}{2}R_3 + \frac{1}{2}R_2} \left[\begin{array}{cc|cc} 1 & 0 & 0 & 2 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \end{array} \right] \xrightarrow{-R_2 + R_3} \left[\begin{array}{cc|cc} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & -1 \end{array} \right]$$

$$\begin{bmatrix} 2 & 3 \\ -3 & 1 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 18 \\ -5 \\ 2 \end{bmatrix}$$

3) Para la matriz $A = \begin{bmatrix} 1 & 2 & -3 \\ 1 & -2 & 1 \\ 1 & -2 & 1 \end{bmatrix}$ determine si existe una matriz invertible P tal que diagonalice a la matriz.

$$-\lambda^3 + \text{traza } \lambda^2 - M_{ii} \lambda + \det(A)$$

$$\text{traza} = 0$$

$$M_{11} = \begin{vmatrix} -2 & 1 \\ -2 & 1 \end{vmatrix} = -2 - (-2) = -2 + 2 = 0$$

$$M_{22} = \begin{vmatrix} 1 & -3 \\ 1 & 1 \end{vmatrix} = 1 - (-3) = 1 + 3 = 4$$

$$M_{33} = \begin{vmatrix} 1 & 1 \\ 1 & -2 \end{vmatrix} = -2 - (1) = -2 - 1 = -3$$

$$M_{ii} = 0$$

$$\det(A) = 1 \begin{vmatrix} -2 & 1 \\ -2 & 1 \end{vmatrix} = 0 \quad 2 \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} = 2(0) = 0$$

$$-3 \begin{vmatrix} 1 & -2 \\ 1 & -2 \end{vmatrix} = -3(0) = 0$$

$$-\lambda^3$$

$$\text{Valor propio} = 0$$

$$\begin{bmatrix} 1 & 2 & -3 & | & 0 \\ 1 & -2 & 1 & | & 0 \\ 1 & -2 & 1 & | & 0 \end{bmatrix} \xrightarrow{\substack{-1F_1 + F_2 \\ -1F_1 + F_3}} \begin{bmatrix} 1 & 2 & -3 & | & 0 \\ 0 & -4 & 4 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \xrightarrow{-\frac{1}{4}F_2} \begin{bmatrix} 1 & 2 & -3 & | & 0 \\ 0 & 1 & -1 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

$$\begin{aligned} x - z &= 0 \\ y - z &= 0 & z = z \\ x &= z \\ y &= z \end{aligned}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} z \\ z \\ z \end{bmatrix} = z \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

no es diagonalizable, pues su MA no es igual a su geométrica

4) muestre que la T.L. $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ definida por la expresión:

$$T[(x, y)] = (x + y, 2x + 3y) \text{ es invertible y } T^{-1}[(1, 1)]$$

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix} \quad \det = 3 - 2 = 1$$

$$A^{-1} = \begin{bmatrix} 3 & -1 \\ -2 & 1 \end{bmatrix} = (3x - y; -2x + y) \quad x = 1 \quad y = 1$$

$$\begin{bmatrix} 3 & -1 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

b) Verif. que que $x = \begin{bmatrix} 2 \\ 6 \\ 1 \end{bmatrix}$ es un v.t.p de $M = \begin{bmatrix} 4 & -1 & 2 \\ -6 & 5 & -6 \\ -6 & 3 & -4 \end{bmatrix}$ y determine el v.l.p correspondiente.

$$\begin{bmatrix} 4 & -1 & 2 \\ -6 & 5 & -6 \\ -6 & 3 & -4 \end{bmatrix} \begin{bmatrix} 2 \\ 6 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 12 \\ 2 \end{bmatrix} = 2 \begin{bmatrix} 2 \\ 6 \\ 1 \end{bmatrix}$$

c) Hallar $P = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$ $D = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$, calcule A^3

$$D^3 = A^3 = \begin{bmatrix} 1 & 0 \\ 0 & 27 \end{bmatrix}$$